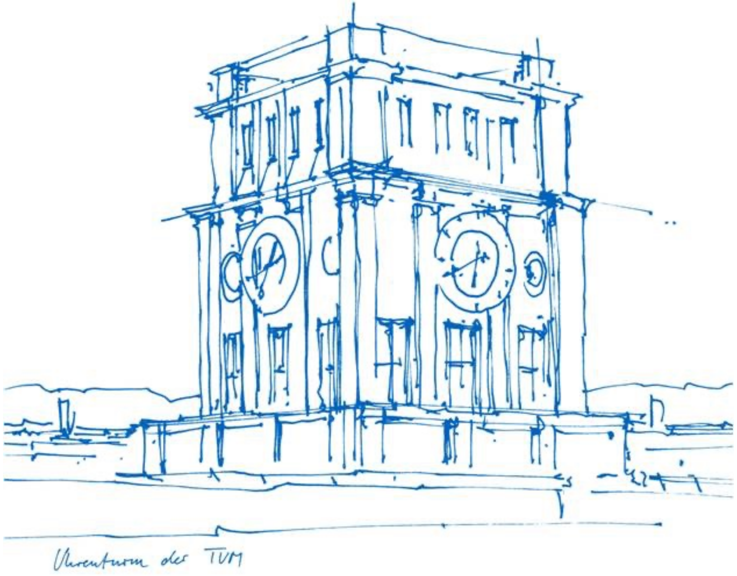
Length-of-Stay (LOS) prediction in hospitals and managerial analysis of hospital data

- Final Project -

Programming for Management Studies (MGTHN0144)

Bachelor in Management and Data Science (B.Sc.)

Heilbronn, 16.07.2026



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Structure

- Our dataset
- Managerial aspects and questions to be answered
- Preparation of the data
- Answering the given question
- ML model to predict LOS
- Conclusion
- Contributions of Team Members

Our dataset

- From Kaggle (.csv)
- Rows: 313,793 (after cleaning)
- Columns: 18

Feature	**Data Type**
case_id	Numeric
Hospital_code	Numeric
Hospital_type_code	Categorical (Text)
City_Code_Hospital	Numeric
Hospital_region_code	Categorical (Text)
Available Extra Rooms in Hospital	Numeric
Department	Categorical (Text)
Ward_Type	Categorical (Text)
Ward_Facility_Code	Categorical (Text)
Bed Grade	Numeric
patientid	Numeric
City_Code_Patient	Numeric
Type of Admission	Categorical (Text)
Severity of Illness	Categorical (Text)
Visitors with Patient	Numeric
Age	Categorical (Text)
Admission_Deposit	Numeric

Limitations of dataset

- Missing values in datasets -> data cleaning!
- Unevenly distributed amongst LOS categories
- Columns not clearly defined/specified
  - city_code
  - region_code
  - hospital_type

Managerial aspects and questions to be answered

(1)

Which department produces the longest lengths of stay and is thus the most costly for hospitals?

-> Solution via aggregation and statistical analysis of the dataset

(2)

Length of Stay (LOS) prediction for a new patient

-> Solution via Training and Implementation of own Machine Learning (ML) Prediction Model

Preparation of the given data

Data Cleaning

I. rows that miss at least one value are dropped

II. changing the case_id s.t. it is continuous again after deleting rows

III. overwriting the .csv-file with the new file

➔ from 455,495 to 313,793 rows (-31.11%)

(1) Which department produces the longest lengths of stay and is thus the most costly for a hospital?

Abstract (I)

- Financing in German healthcare is based on G-DRG system
  - financing time window assigned for different types of diagnoses
- for hospitalizations exceeding the threshold („obere Grenzverweildauer"):

  hospital receives only a flat per diem payment!

- long hospital stays: usually tied to severe cases -> patients often require specialized care → greater share of resources.

- **Increased LOS → Increased financial burden!**

Abstract (II)

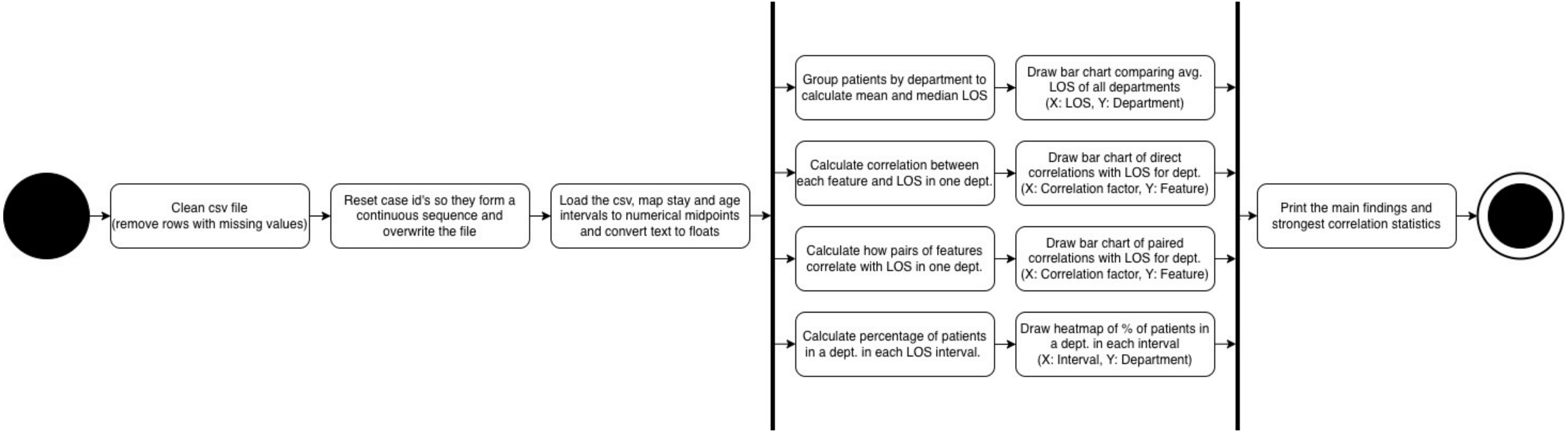
- actionable steps: identify the specific departments dealing with prolonged stays most often
- allocating resources to improve a department's efficiency: Investments in…
  - …better and newer equipment,
  - …increasing human resources,
  - …hiring specialists

→ **all decided based on which department is struggling the most!**

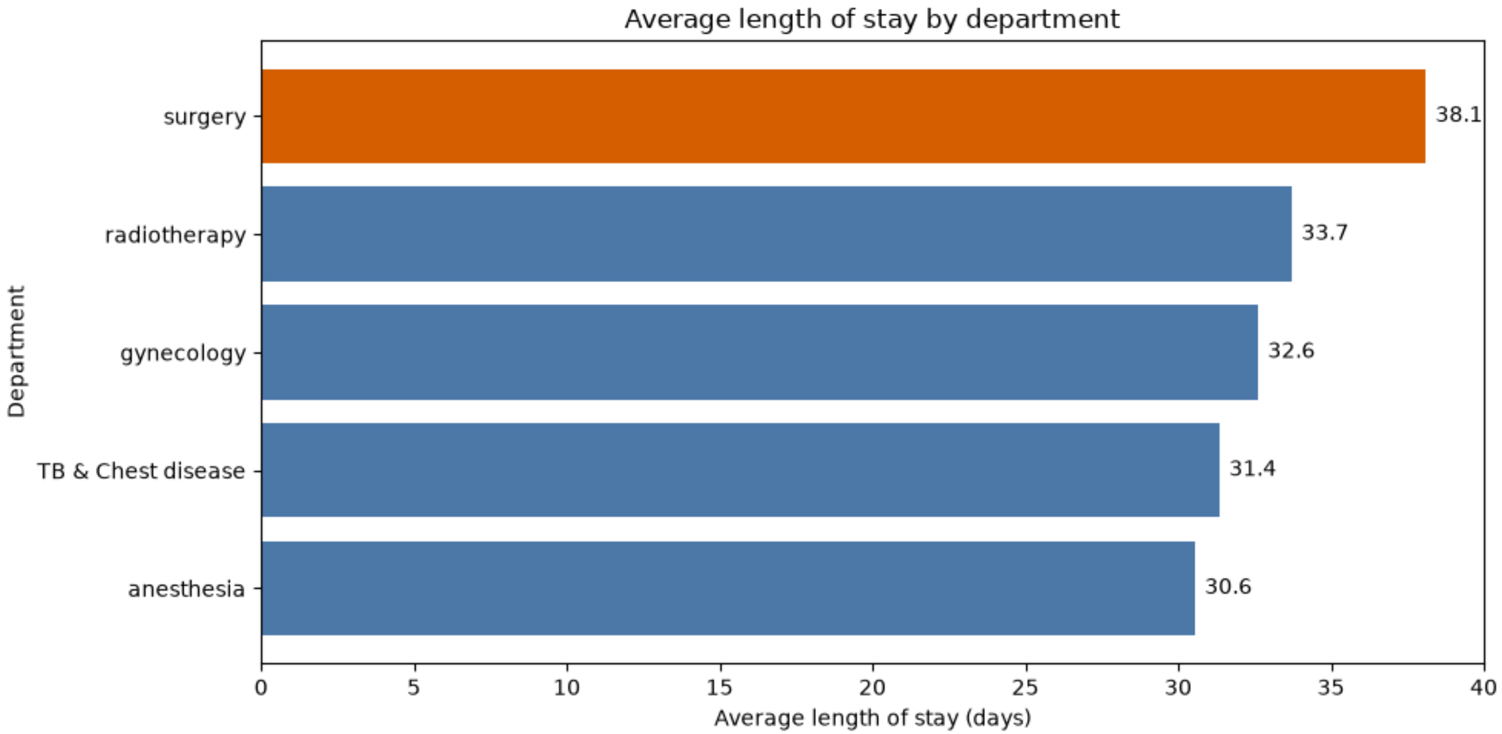
Abstract (III)

Goal: analyzing the provided data, we aim to find which department, on average, struggles with long Lengths of Stay (LOS) across hospitals and would therefore be the best candidate for a capital investment.

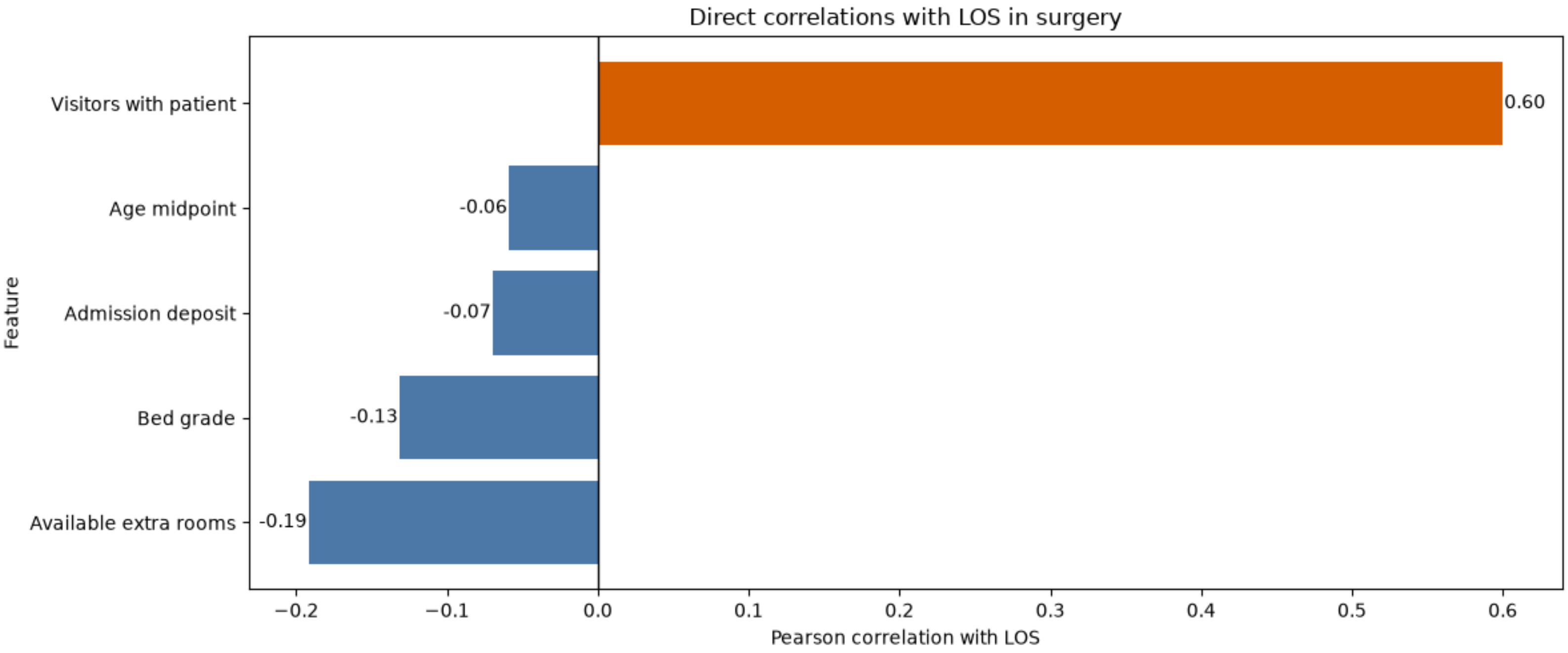
Process of our analysis (UML)



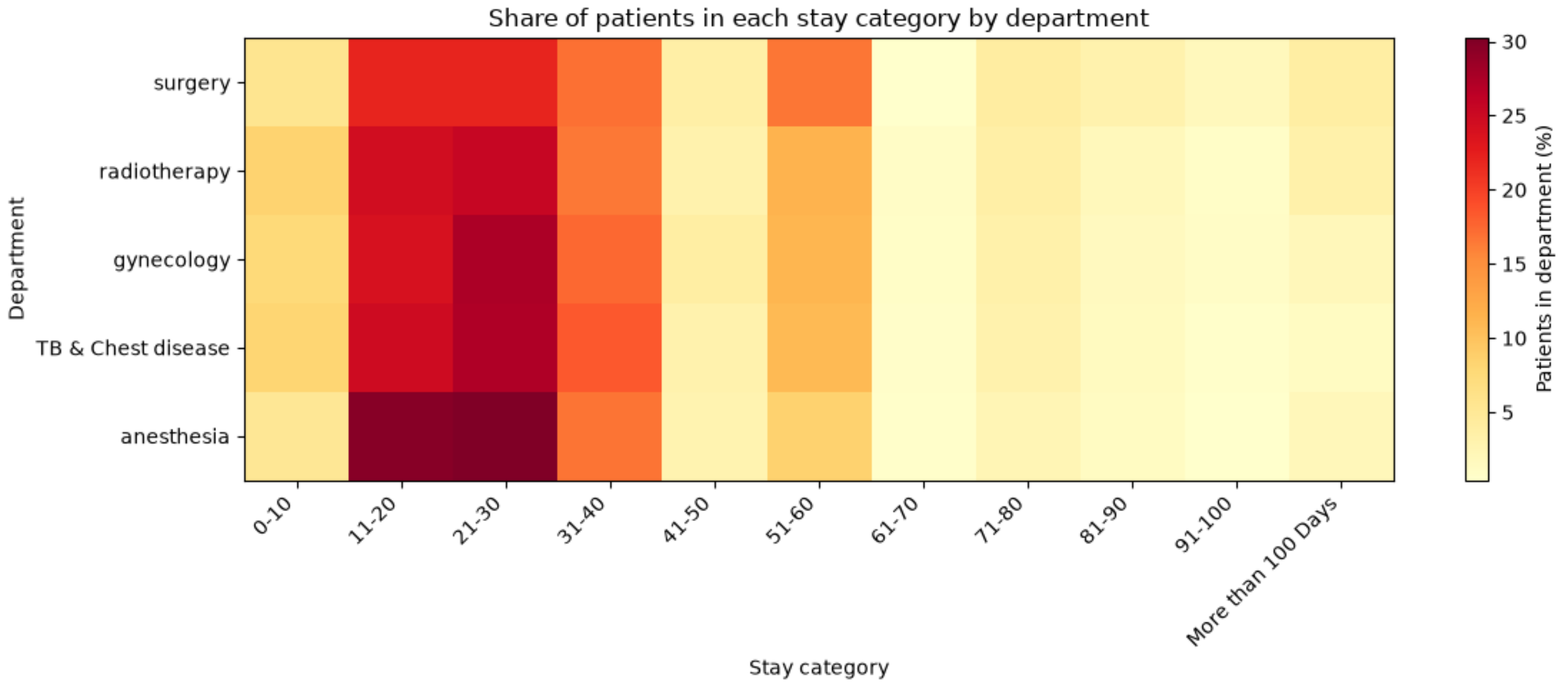
Statistical findings



→ The **surgical department** has the longest average length of stay with 38.1 days



→ Inside surgery, the strongest direct correlation with LOS is Visitors with patients (r = 0.60)



(Surgical department: **least cases across data** (1,143 patients against e.g. 245,850 in gynecology))

Managerial advice for hospitals

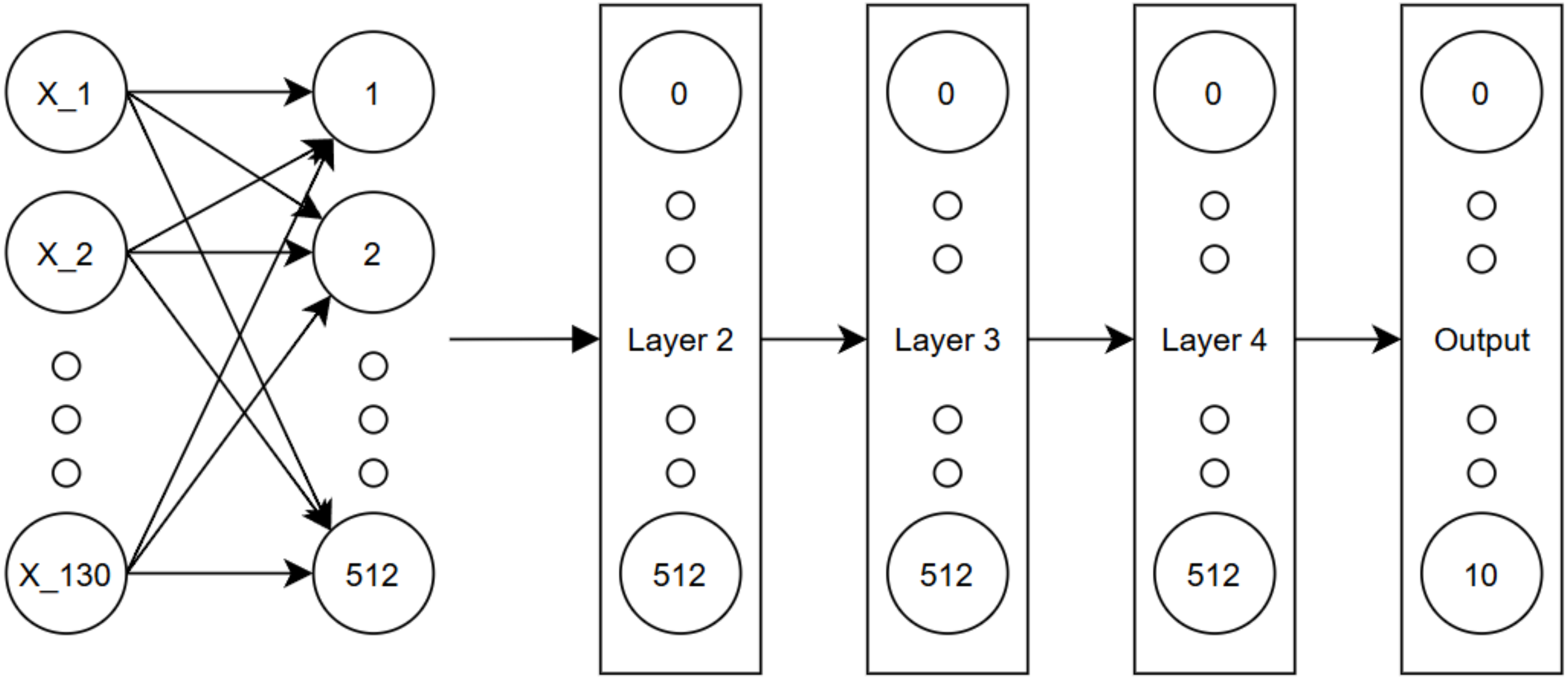
- Surgery:
  - longest average LOS, thus: prime candidate for efficiency investment.
  - least cases -> struggles most per patient, but limited absolute volume of costs. (Still crucial when reducing economic losses under DRG system)
- Within surgical departments: number of visitors with the patient shows by far the strongest correlation with LOS -> indicating case severity and required care intensity
  - Observable early in the stay and without further analysis
  - Can serve as early-warning indicator for cases which risk exceeding the DRG time window ("obere Grenzverweildauer")

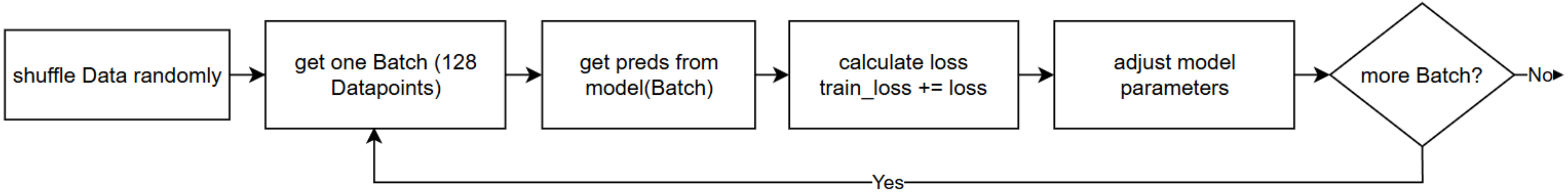
→ recommended actions: increase direct investment into the surgical department and use a high visitor count during the stay as a possible indication for exceeding the allocated time window.

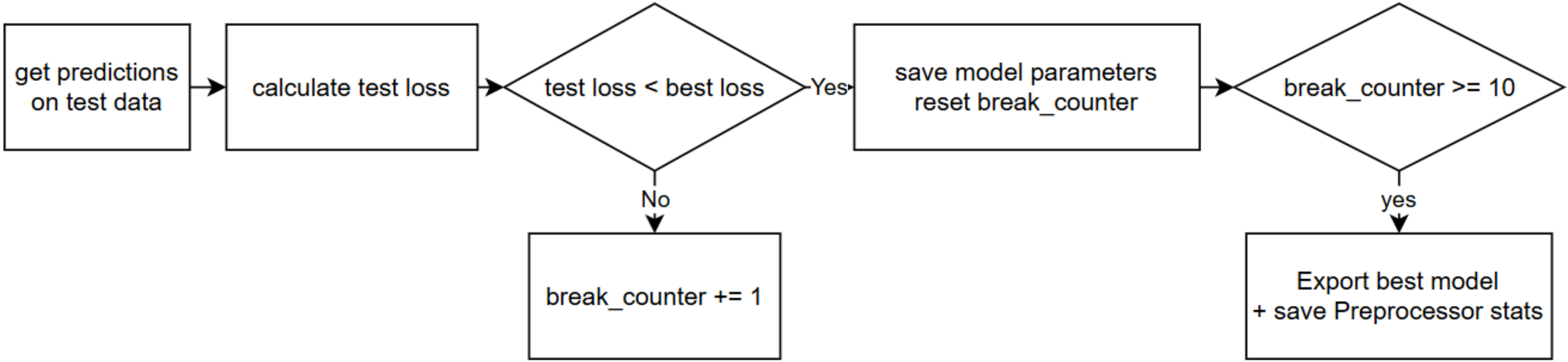
(2) Length of stay (LOS) prediction for a new patient via self-trained Machine Learning (ML) model

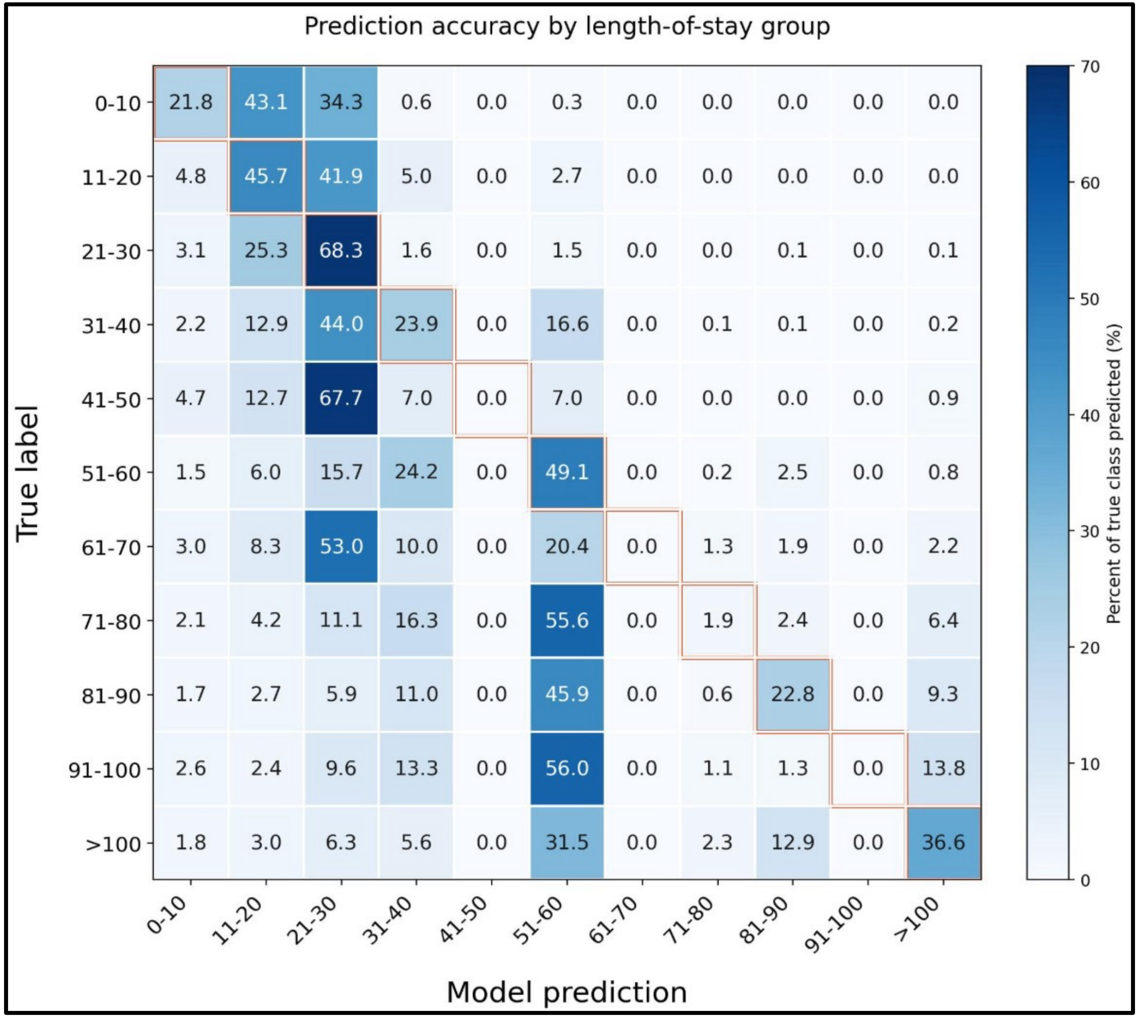
Overview

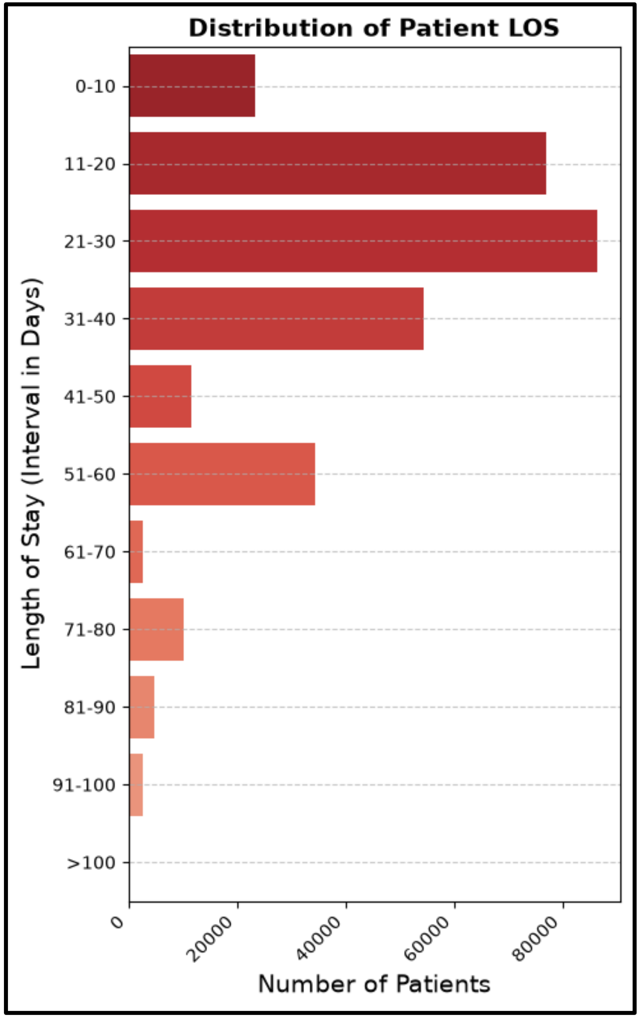
- Architecture
  - 6-layer MLP
  - Pytorch
  - Adam (optimizer)
  - Loss function: Cross-entropy-loss


X_1
X_2
X_130
1
2
512
0
Layer 2
512
0
Layer 3
512
0
Layer 4
512
0
Output
10


shuffle Data randomly
get one Batch (128 Datapoints)
get preds from model(Batch)
calculate loss
train_loss += loss
adjust model parameters
more Batch?
No
Yes


get predictions
on test data
calculate test loss
test loss < best loss
Yes
save model parameters
reset break_counter
break_counter >= 10
No
break_counter += 1
yes
Export best model
+ save Preprocessor stats


Prediction accuracy by length-of-stay group
True label
Model prediction
Percent of true class predicted (%)
0-10: 21.8 43.1 34.3 0.6 0.0 0.3 0.0 0.0 0.0 0.0 0.0
11-20: 4.8 45.7 41.9 5.0 0.0 2.7 0.0 0.0 0.0 0.0 0.0
21-30: 3.1 25.3 68.3 1.6 0.0 1.5 0.0 0.0 0.1 0.0 0.1
31-40: 2.2 12.9 44.0 23.9 0.0 16.6 0.0 0.1 0.1 0.0 0.2
41-50: 4.7 12.7 67.7 7.0 0.0 7.0 0.0 0.0 0.0 0.0 0.9
51-60: 1.5 6.0 15.7 24.2 0.0 49.1 0.0 0.2 2.5 0.0 0.8
61-70: 3.0 8.3 53.0 10.0 0.0 20.4 0.0 1.3 1.9 0.0 2.2
71-80: 2.1 4.2 11.1 16.3 0.0 55.6 0.0 1.9 2.4 0.0 6.4
81-90: 1.7 2.7 5.9 11.0 0.0 45.9 0.0 0.6 22.8 0.0 9.3
91-100: 2.6 2.4 9.6 13.3 0.0 56.0 0.0 1.1 1.3 0.0 13.8
>100: 1.8 3.0 6.3 5.6 0.0 31.5 0.0 2.3 12.9 0.0 36.6
0-10 11-20 21-30 31-40 41-50 51-60 61-70 71-80 81-90 91-100 >100
0 10 20 30 40 50 60 70


Distribution of Patient LOS
Length of Stay (Interval in Days)
Number of Patients
0-10
11-20
21-30
31-40
41-50
51-60
61-70
71-80
81-90
91-100
>100
0 20000 40000 60000 80000

Conclusion and limitations (I)

- Some classes (e.g. [41-50]): predicted with 0% accuracy
  - Only few rows of training data available for that class
  - During training: model optimized for overall loss across all samples
  - -> underrepresented class: model learns: statistically safer to predict a majority class instead of risking a wrong guess on a rare one!
  - E.g.: [21-30] has by far the most training examples
    - model learns a high prior probability, tends to default to predicting [21-30]
    - even for inputs that belong to a minority class, e.g. [41-50].

  - **Result: model essentially "gives up" on the rare classes in favor of minimizing loss on the common classes!**

Conclusion and limitations (II)

- **Model can still be used as rough estimate/predictor**

- **Model could be refined through collection and usage of more training data.**
- **Infrastructure of the model is suitable!**

Final conclusion

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- **Analysis of 313,793 cleaned hospital records.**
- **Surgical department: longest average LOS** (38.1 days, median 35.5)

- Within surgical department: visitor count correlates most strongly with LOS (r = 0.60)
  - -> cheap early-warning indicator for stays at risk of exceeding the DRG time window
- Surgical department: efficiency investment suitable, though high-volume gynecology (245,850 cases) carries the largest absolute exposure.
- Self-trained machine learning model: prediction of a newly admitted patient's LOS category from admission-time features
  - **42.3% test accuracy across 11 classes**
  - accurate enough to support e.g. bed capacity planning, early discharge management, and staffing decisions at admission.

Thank you for your time and attention!

Contributions of Team Members

- Muhammad Sheheryaar Ghayas (03801683): Creation of UML diagrams, explorative analysis of data, creation of further visualisations and functions
- Vsevolods Mitrevics (03815799): Creating functions for data loading, cleaning, aggregating and visualizing (for Task I).
- Constantin Demuth (03818527): Creating, training, testing and documenting ML model (for Task II).
- Stanislaw Pohl (03822985): Data analysis and exploration regarding suitable management questions, visualisations, research on healthcare system and contextual information
- Jannis Nickel (03813516): Writing report, creationg presentation slides, Explorative Data analysis, Github Repo Management, Compilation of final notebook & deliverables

(Please note that we spent a considerable amount of effort on exploring the given data before making a decision on a managerial aspect we want to analyze for our project. Thus, contributions in the final deliverables may not look equally distributed.)